

Blockchain-Enabled Distributed Authentication Mechanism for Industrial Device Access

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Abstract—The secure access of numerous heterogeneous devices ensures the stability of the industrial Internet of Things. Centralized authentication can be overwhelmed by an influx of authentication requests from malicious devices. Distributed device-to-device authentication is vulnerable to tampering attack evidence. Blockchain authentication eases evidence tampering, but traditional blockchains impose high-performance requirements on devices, rendering them unsuitable for resource-limited devices. Therefore, in this article, we apply IOTA to device access authentication. To the best of authors' knowledge, this is the first application. Based on this, we propose a blockchain-enabled distributed authentication mechanism for industrial device access. The authentication involves a consensus phase based on optimized IOTA and a unique identification code validation phase. We optimized the tip selection algorithm of IOTA to make it faster and more stable. The performance experiment results demonstrate that our mechanism meets the time-consuming requirements of device access. The security experiments indicate that the authentication phases effectively intercept the attacks.

Index Terms—Blockchain, device access, industrial Internet of Things (IIoT), lightweight distributed authentication.

NOMENCLATURE

Notations and Their Descriptions

T^{ED}	Transaction of edge device.
T^{FD}	Transaction of field device.
TT	Tip transaction.
W	Cumulative weight of a transaction.
ε	Step size generated by the Lévy flight algorithm in (4).
α	Parameter of MR.
θ	Parameter of LFMR.
K	Number of iterations.
Z_k	Number of data sets for different α and θ in the k th iteration.
N_j	Number of data sets for different node numbers in the j th set of data in Z .

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$\bar{x}_{jk}^L$	Average number of nodes selected by LFMR calculated by the j th set of data in Z_k .
$\bar{x}_{jk}^M$	Average number of nodes selected by MR calculated by the j th set of data in Z_k .
x_{ij}	Number of nodes found for the i th set of data in N_j .
y_{ij}	Number of nodes found for the i th set of data in N_j .
CL	Confidence level.

I. INTRODUCTION

THE interconnection of numerous industrial field devices (FDs) is the base of the industrial Internet of Things (IIoT). The devices vary from microcomputers, actuators, and sensors [1], whose manufacturers include Siemens, NXP semiconductors, and Schneider Electric [2]. IIoT systems access these devices in several manners, such as hardware resource reconfiguration, physical interface expansion, and dynamic program generation through edge-cloud collaboration [3]. The networks used to connect them are also diverse [4]. Wired network technologies such as process field net (PROFINET), EtherCAT, SERCOS III, and EtherNet/IP [5] are usually used to realize real-time data acquisition and high-precision control. Wireless communication technologies, including ZigBee, Z-Wave, Wi-Fi, LoRa, and 5G network, are contributing to the variety and number of accessible devices and expanding the applications of IIoT [6].

The heterogeneity of devices, along with the diversity of access methods and networks, increases the complexity of IIoT systems, thereby augmenting the probability of security issues [7]. The security issues in IIoT coverage from the device layer to the platform layer [8], leading to severe consequences in the petrochemical, power [9], and nuclear industries [10]. Compared to other layers, the device layer has millions and billions of devices, distributed and unattended. Attackers are more likely to launch attacks from the device layer by creating malicious devices [11], [12]. Therefore, we intend to start our research on the device layer, designing a universally applicable security technique to enhance the security of IIoT systems. Considering the limited resources of numerous devices, rather than incorporating security mechanisms directly into each device, our research focuses on device access authentication to prevent malicious devices from attacking IIoT systems. As shown in Fig. 1, current research on device access authentication primarily focuses on centralized authentication, device-to-device authentication, and blockchain-based authentication.

Centralized authentication: This solution uses a central authentication service to authenticate accessed devices. Gamer et al. [13] protected the data confidentiality and privacy for IoT using an OPC-UA-based authentication mechanism, and the protocol used is the lightweight directory access protocol. The authorization is executed by a Center Authservice Server. Liang

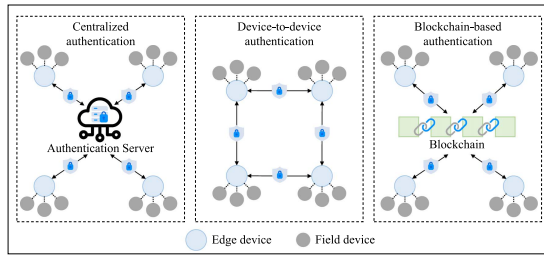


Fig. 1. Current primary research schemes for device access authentication.

et al. [14] proposed a two-way IC authentication protocol based on physical unclonable functions (PUF) for low-power radio frequency identification (RFID) systems. This protocol allows two-way authentication between a server and an RFID tag. The reliance on centralized service for device access authentication can pose significant risks. An influx of authentication requests from malicious devices can overwhelm the centralized service, impeding the authentication process for legitimate devices. Therefore, the researchers have proposed device-to-device authentication schemes to decentralize the authentication requests.

Device-to-device authentication: This is a distributed authentication approach that performs group authentication or mutual authentication between devices at the device layer. Zhao et al. [15] proposed a group authentication scheme that utilizes aggregate authentication technology to authenticate groups of narrowband Internet of Things (NB-IoT) devices. Adil et al. [16] proposed a device-to-device authentication scheme for cyber-physical systems. It employs a secured authentication key generated from a hash function with a MAC address to verify the legitimacy of requesting devices. The key is published to the other legitimate devices through the base station for one-way authentication. Meier et al. [17] proposed a portable trust anchor concept for OPC unified architecture (OPC-UA). It authenticates the OPC-UA clients before connecting to the main global discovery server (GDS). Dang et al. [18] proposed a hybrid approach where the servers and the devices are in charge of different security certification stages through an authentication protocol. The servers are responsible for managing access rights and maintaining a list of the things in the IoT local network. The responsibility of the devices is to perform device-to-device authentication. However, device-to-device authentication has certain limitations concerning flexibility and scalability. Hence, blockchain technology is adopted [19].

Blockchain-based authentication: Blockchain technology can address the problem of data tampering [20]. Cui et al. [21] designed a blockchain-empowered device cross-domain authentication scheme to achieve secure collaboration among devices in different domains. Shen et al. [22] proposed an anonymous device authentication approach supported by blockchain technology to implement mutual authentication of devices located in different IIoT domains. Zhang et al. [23] presented a cross-domain device authentication method that integrates blockchain and multifactor authentication. The blockchain is utilized to store the dynamic accumulator, which assists in verifying the legitimacy of devices in each domain. In industrial scenarios, a device should be authenticated within an acceptable time horizon, or it will be unable to transfer data to the IIoT system on time. Traditional blockchains cannot accommodate the requirements of IIoT for latency, throughput, and network extensibility. In particular, traditional blockchains require more computation and storage resources as the number of devices

grows [24]. Therefore, the current application of blockchain in device access authentication mainly focuses on authentication when devices collaborate across IIoT domains. Instead of deploying blockchain on every IIoT device, an additional server is set up in each domain to handle all blockchain authentication procedures. However, each IIoT device has a certain degree of computing power. A lightweight blockchain is needed to fully leverage the computing power of these devices for device authentication and eliminate the necessity for additional servers.

IOTA is an ideal solution for IIoT device access security authentication. Compared to conventional blockchains, the ledger of IOTA, IOTA tangle (Tangle), is implemented based on DAG. This structure of Tangle allows for the attachment of unlimited transactions, improving scalability [25]. The distributed feature also makes the system highly scalable, allowing it to accommodate the growing amount of IIoT devices. IOTA supports high-frequency microtransactions, which increases the throughput, making it well-suited for the requirements of the IIoT scenario where numerous devices continuously send authentication transactions [26]. IOTA's distributed consensus and lightweight proof-of-work features significantly reduce computing requirements, allowing it to quickly confirm transactions when running on resource-constrained IIoT devices, avoiding waiting a long time during device access authentication [27], [28]. Thus, IOTA can be applied to the access authentication of a large number of devices. IOTA has been proven to work well on IIoT devices [29], but it is currently used primarily to secure data sharing and has not been applied to secure device access [30], [31].

In addition, regarding device access, its tip selection algorithm may lead to prolonged waiting time for some devices. The design of IOTA omits the role of miners. Each IOTA node serves as both the initiator and confirmer of transactions. Each time an IOTA node initiates a transaction request, it must confirm two unconfirmed transactions within the Tangle, known as tip transactions (TTs). The tip selection algorithm is employed to select two TTs from the Tangle. This algorithm generates two random walk paths within the Tangle and iteratively traverses these paths until two TTs are found. It enhances IOTA's resilience against double spending attacks [26]. Furthermore, the impact of this algorithm can be tuned through a parameter. However, adjusting the parameter requires a tradeoff between security and the universality of TT confirmation. Enhancing security may lead to a situation where transactions initiated by numerous devices will not be confirmed, but prioritizing the universality of TT confirmation might compromise security. Striking a balance between security and universality becomes crucial.

Therefore, we propose an authentication mechanism based on optimized IOTA for device access industrial scenarios. In summary, we have made the following contributions.

- 1) A lightweight authentication mechanism for resource-limited device access in distributed industrial scenarios is proposed. It contains a consensus phase based on optimized IOTA and a unique identification code (UIC) validation phase. The IOTA is a lightweight blockchain and is used to intercept attacks, thus alleviating the processing pressure in the second phase. In addition, to the best of our knowledge, this is the first application of IOTA in IIoT device access authentication.
- 2) We propose an optimized tip selection algorithm incorporating the hopping mechanism of Lévy Flight algorithm to dynamically adjust the α value of the Markov Chain Monte Carlo random walk algorithm (MR). We conduct

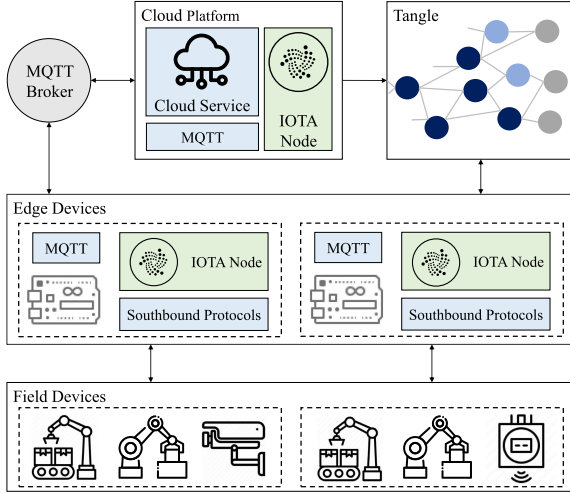


Fig. 2. Overview of the proposed architecture. The EDs connect the FDs and initiate authentication requests to the Tangle. The Tangle is used to manage authentication request messages. The cloud platform is responsible for interacting with the devices and obtaining their operating data.

comparative experiments between the two algorithms. Experimental results demonstrate that the optimized algorithm achieves better results in tip selection speed, stability, and transaction confirmation efficiency.

- 3) We conduct performance experiments, and the results indicate that our proposed mechanism meets the authentication time and system throughput requirements of IIoT scenarios. We also conduct security experiments, and the results show that the mechanism effectively avoids numerous legitimate devices from facing authentication challenges caused by malicious attacks.

The rest of this article is organized as follows. Section II describes the system models of our mechanism. Section III demonstrates the implementation details of the proposed solution. Section IV depicts how the mechanism was evaluated. Finally, Section V concludes this article.

II. SYSTEM MODEL

The proposed blockchain-enabled distributed authentication mechanism utilizes the following system architecture and authentication model.

A. System Architecture

Fig. 2 depicts an access architecture for IIoT devices. This architecture contains five categories of entities: FD, edge device (ED), message queuing telemetry transport (MQTT) broker, Tangle, and cloud platform. FDs are heterogeneous devices, such as industrial equipment, robots, cameras, and sensors. The protocols the FDs use for transferring data and control commands with EDs differ from their types, models, or even versions, such as Profibus, OPC-UA, controller area network bus (CANBus), and real time streaming protocol (RTSP). The EDs principally contain embedded EDs with limited resources and are responsible for accessing FDs and collecting data from them. An ED contains a southbound protocol module, an IOTA node module, and an MQTT module. It can connect to and manage multiple FDs. The southbound protocol module within it interacts with the FDs utilizing their respective communication

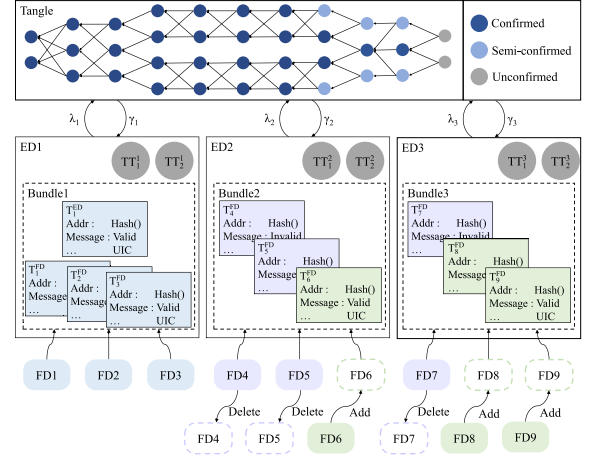


Fig. 3. Blockchain-based consensus. The ED1 shows a new ED performs authentication operations for itself and the new FDs it connects. The ED2 and ED3 show the authentication operations for FD modifications. The Tangle contains three types of nodes: confirmed, semi-confirmed, and unconfirmed.

protocols. This interaction involves gathering information and data from the FDs while also transmitting control commands to them. The IOTA node module communicates with the Tangle, sending authentication transactions to it or acquiring TTs from it. The MQTT module within it, along with the one within the cloud platform, establishes a data interaction channel between the two entities through the MQTT broker. The MQTT broker, serving as a message-oriented middleware, is employed in this architecture to transfer the data between EDs and the cloud platform. To join the system, an ED must initiate an authentication request to the Tangle and submit registration requests to the cloud platform for both itself and the connected FDs. Once authenticated, they can become members of the system. The Tangle is a ledger built using DAGs. It enables data exchange between IOTA nodes and records historical transactions. The cloud platform contains a cloud service module, an MQTT module, and an IOTA node module. It authenticates the legitimacy of EDs and interacts with them once their legitimacy is confirmed. The MQTT module communicates with EDs, receiving their registration requests and exchanging data with them. The IOTA node module retrieves device information from Tangle and employs it to verify the legitimacy of the devices.

B. Device Authentication Model

The authentication of devices includes two phases, the blockchain-based consensus and the UIC validation. The blockchain-based consensus is shown in Fig. 3. It uses IOTA with an optimized tip selection algorithm to guarantee the reasonableness of the devices. Aside from improving the cost of the attack, since the transactions generated by the attackers will also become part of the Tangle, this part also preserves the information of attackers. ED1, FD1, FD2, and FD3 are new devices. ED2 and ED3 are existing devices. FD4, FD5, and FD7 are devices to be deleted. FD6, FD8, and FD9 are devices newly connected to existing EDs. $\text{Bundle}_i = \{T_i^{\text{ED}}, T_j^{\text{FD}}, T_{j+1}^{\text{FD}}, \dots, T_k^{\text{FD}}, \text{TT}_1^i, \text{TT}_2^i\}$ is a set of authentication requests. T_i^{ED} is a transaction of ED_i, and $T_j^{\text{FD}}, T_{j+1}^{\text{FD}}, \dots, T_k^{\text{FD}}$ are transactions of FDs that ED_i connects. γ_i is a network packet from the Tangle to ED_i, which carries two TTs TT_1^i and TT_2^i . λ_i is a network packet that used to send Bundle_i from ED_i to the Tangle. As described in the

message field, both T_i^{ED} and T_j^{FD} have two types. The “Valid” value in the message field means adding a new device, while the “Invalid” value means deleting an existing device. After this authentication process is completed, the TT_1^i and TT_2^i will turn into semiconfirmed nodes, and the $Bundle_i$ will become a new unconfirmed node in the Tangle.

Given that the IIoT has real-time data transmission requirements, using the Tangle to transmit data will result in some latency. Accordingly, this article adopts UIC for data interaction authentication between cloud platforms and devices. The UIC validation is used to guarantee the legitimacy of the devices. In this part, the devices are described using a 5-tuple $M = \{DF, DT, DV, DN, DA\}$. DF is the device manufacturer, DT is the device model, DV is the device version, DN is the device serial number, and DA is the address generated from a seed and required by each transaction propagated in IOTA. In every ED, a seed is required, and the seed works for both the ED and the FDs it connects to. Each device belonging to FDs or EDs generates a UIC according to the following equation:

$$UIC = \mathcal{H}(\kappa \oplus opad | \mathcal{H}(\kappa \oplus ipad | M)) \quad (1)$$

where $\mathcal{H}$ is a hash function; κ is the secret key generated according to DA; $opad$ and $ipad$ are fixed strings with the same length as κ . The validation is performed by the cloud platform. In the UIC validation phase, the cloud platform periodically retrieves unique identifiers of the newly confirmed transactions from the Tangle, such as the addresses of the transactions. The unique identifiers are leveraged to create MQTT topics for device registration. The cloud platform will then subscribe to these topics via the MQTT broker. Each topic is used only once and is unsubscribed following the processing of the corresponding device registration request. A device will publish a registration request to the topic the cloud platform subscribes via the MQTT broker, carrying its attribute information. To verify the legitimacy of a device, the cloud platform utilizes the DA of the device to retrieve the corresponding transaction from the Tangle. Then, it extracts the UIC from the message field of the transaction. Meanwhile, the cloud platform obtains the pre-entered UIC' from its database according to the DF, DT, DV, and DN of the device. A comparison is then conducted to ascertain the equality between the UIC and the UIC' . The same value of UIC and UIC' means that the device is legal, and vice versa.

C. Threat Model

We consider the threat posed by adversaries aiming to compromise the device access authentication process and disrupt the registration service of the IIoT server, a crucial security requirement for device-server interactions. The adversary's primary objective is to covertly circumvent the security authentication mechanism, thereby gaining access to sensitive data within the IIoT system. In addition, by flooding the registration service with frequent requests, the adversary can disrupt legal device registrations, resulting in numerous IIoT devices failing to work correctly and raising the risk of system paralysis. We assume that, within our proposed mechanism, entities engage in data exchanges via secure channels, and we exclude the case where an adversary implants malware into IIoT devices to steal real-time statuses and control the devices for attack purposes.

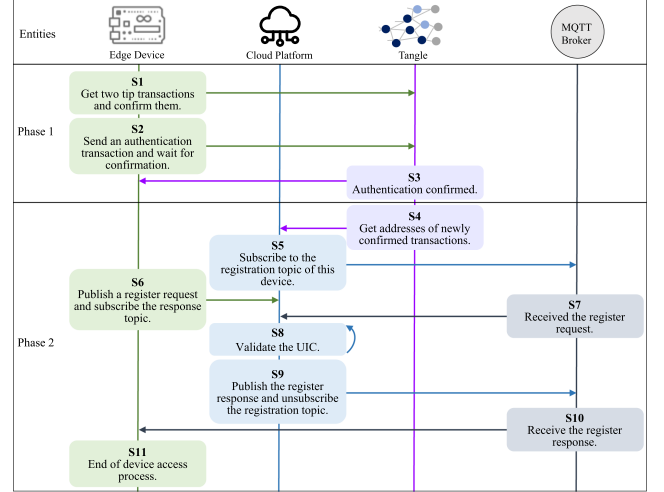


Fig. 4. System workflow. There are 11 steps, from S1 to S11. Phase 1 represents the blockchain-based consensus phase, while Phase 2 represents the UIC validation phase.

III. PROPOSED MECHANISM

The system workflow is depicted in Fig. 4, encompassing primarily 10 steps.

A. FD Access

FDs are heterogeneous devices with different communication protocols and data formats. Furthermore, the data that requires collection from a particular device can differ from scenario to scenario. Thus, we proposed a dynamic FD access method. The communication protocol of an FD is described using a 3-tuple $P = \{PK, PI, PIN\}$. PK is the protocol index key used to retrieve specific protocols, PI is the FD's physical interface, and PIN is the FD's physical interface number. PK is calculated from (2). DF, DT, DV, PI, and PIN are configured in the ED that the FDs connect to when deploying. Then, the ED turns DF, DT, and DV into numerical digits according to the search table defined in advance. The numerical digit sizes of the three elements are μ_1 bits, μ_2 bits, and μ_3 bits each

$$PK = DF * 2^{\mu_2 + \mu_3} + DT * 2^{\mu_3} + DV. \quad (2)$$

The ED sends a protocol parsing program generation request containing PK to the cloud platform to generate the program automatically.

B. ED Registration

The registration is to verify the legitimacy of the EDs. Before sending a registration request to the cloud platform, the reasonableness of the device should be validated through Algorithm 1. To initiate a device reasonableness validation request, an ED should create a transaction T^{ED} for itself and transactions T^{FDs} for the FDs it connects based on the device attributes and status. A Valid ED can create both Valid and Invalid transactions for the FDs it connects, and an Invalid ED can only create Invalid transactions. These transactions will be packaged into a Bundle. Before sending the Bundle to the Tangle, the ED obtains two TTs $\{TT_1, TT_2\}$ from the Tangle to confirm them. The confirmation procedure is:

Algorithm 1: Device Reasonableness Validation.

```

1: Input:  $ED, FDs$ 
2:  $UIC = \mathcal{H}(\kappa \oplus opad | \mathcal{H}(\kappa \oplus ipad | ED.M))$ ;
3: if  $ED.Status == Valid$  then
4:    $T^{ED} = NewTrans(UIC, ED.DA, Valid)$ ;
5: else
6:    $T^{ED} = NewTrans(UIC, ED.DA, Invalid)$ ;
7: end if
8: for  $i = 0; i < FDs.count; i++$  do
9:    $UIC = \mathcal{H}(\kappa \oplus opad | \mathcal{H}(\kappa \oplus ipad | FDs[i].M))$ ;
10:  if  $ED.Status == Valid \&$ 
     $FDs[i].Status == Valid$  then
11:     $T^{FDs}[i] = NewTrans(UIC, FDs[i].DA,$ 
12:       $Valid)$ ;
13:  else
14:     $T^{FDs}[i] = NewTrans(UIC, FDs[i].DA,$ 
15:       $Invalid)$ ;
16:  end if
17: end for
18:  $Bundle = NewBundle(T^{ED}, T^{FDs})$ ;
19:  $\{TT_1, TT_2\} = \gamma()$ ;
20: while true do
21:  if  $ConfirmTips(TT_1, TT_2) == true$  then
22:     $break$ ;
23:  end if
24:   $\{TT_1, TT_2\} = \gamma()$ ;
25: end while
26:  $\lambda(Bundle)$ ;

```

- 1) Check the signature of $\{TT_1, TT_2\}$;
- 2) Make sure that $\{TT_1, TT_2\}$ do not conflict with any of the previous transactions;
- 3) Verify the legitimacy of the UICs in $\{TT_1, TT_2\}$.

If the transaction verification fails, the ED will obtain new TTs from the Tangle.

Following the execution of Algorithm 1, if the ED that initiated the reasonableness validation request has the valid status and is a new device, it is necessary to wait for confirmation by other devices before sending a registration request to the cloud platform.

In Algorithm 1, the time complexity of line 2 is $O(1)$. Lines 3 to 7, which construct a new transaction for an ED and do not involve loops, have a constant time complexity of $O(1)$. Lines 8 to 17 construct the new transactions for the FDs, resulting in a time complexity of $O(n)$. Line 18 packages the transactions. Line 19 obtains two TTs, thus, the time complexity is the same as Algorithm 2, which is $O(n^2)$. Lines 20 to 25 introduce a loop based on line 19, resulting in a time complexity of $O(n^3)$. Line 26, which sends the bundle to the Tangle, has a constant time complexity of $O(1)$. In summary, the overall time complexity of this algorithm is $O(n^3)$. This algorithm necessitates additional space to store the transactions corresponding to the ED and its connected FDs. As the space required is proportional to the number of transactions, the space complexity is $O(n)$.

C. Tip Selection Algorithm

Since a device must be confirmed by other devices before it becomes a legitimate device, the time it takes for a device to be confirmed determines its efficiency in joining the IIoT. The tip selection algorithm is employed to select two TTs from the existing set of TTs within the Tangle for confirmation when a

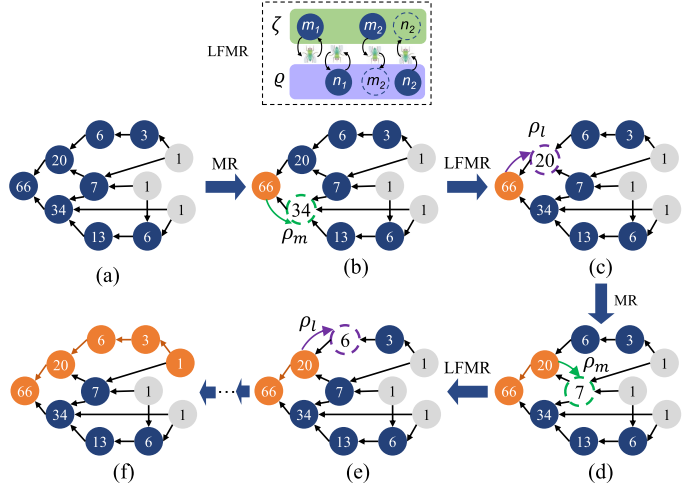


Fig. 5. Working mechanism of the LFMR. The arrows ρ_m denote the paths that MR prefers, and the arrows ρ_l represent the selection of LFMR. In LFMR, ζ represents high-probability space, and η represents low-probability space. m_1 and m_2 represent high-probability transactions, n_1 and n_2 represent low-probability transactions. This figure demonstrates that LFMR can randomly alter the probability of a transaction being selected. (a) is the original DAG. (b) and (d) MR tends to select high-probability transactions. (c) and (e) Transactions selected by LFMR. (f) Final path to TT.

new device is added to the Tangle. Currently, the tip selection algorithm used by the Tangle is MR. The procedure of MR for selecting a single TT involves randomly selecting a non-TT T_i from the existing transactions within the Tangle as the current transaction. Following this, one of T_i 's predecessor transactions, denoted as T_j , is selected based on the probability $P_{T_i \rightarrow T_j}$. Transaction T_j is employed as the new current transaction. This iterative process is repeated until a TT is successfully selected. $P_{T_i \rightarrow T_j}$ is calculated using the following equation:

$$P_{T_i \rightarrow T_j} = \frac{e^{\alpha(W_{T_j} - W_{T_i})}}{\sum_{T_k: T_k \rightsquigarrow T_i} e^{\alpha(W_{T_k} - W_{T_i})}}. \quad (3)$$

T_i is the current transaction, T_k is the set of T_i 's predecessor transactions, and T_j is a transaction among T_k ; W is the cumulative weight of a transaction; α is the parameter of MR, $\alpha > 0$ determines the impact of W on $P_{T_i \rightarrow T_j}$. According to [26], a larger value of α is typically selected to ensure security. However, this may result in some TTs never being confirmed. To optimize the TT selection results of MR, we propose a Lévy flight algorithm optimized MR (LFMR), which can dynamically adjust the value of α to achieve a more balanced TT selection with reduced safety loss.

The working mechanism of LFMR is shown in Fig. 5. Given a large value of α , MR tends to select nodes with smaller $\Delta W = W_{T_j} - W_{T_i}$. Fig. 5(a) shows the original DAG, each node in the DAG has a W . The number displayed on each node represents the node's W . Here, we use W along with its corresponding cumulative weight value to denote a specific node, e.g., W_{66} denotes the node whose W is 66. Fig. 5(b) shows that the tip selection algorithm begins with W_{66} . Upon execution of MR, W_{34} , with a smaller ΔW , tends to be selected from the two predecessor transactions of W_{66} . LFMR may migrate transactions from ζ to η or from η to ζ , resulting in an increase in the possibility of selecting transactions with bigger ΔW values.

Hence, in Fig. 5(c), W_{20} is actually selected. Fig. 5(d) illustrates the selection of W_7 from W_{20} and W_{34} . Fig. 5(e) illustrates the selection of W_1 from W_7 and W_{34} . Fig. 5(f) illustrates the final path to TT.

Algorithm 2: The Transaction Selection Procedure of LFMR.

```

1: Input:  $T_i$ 
2:  $\tau = \text{Random}()$ ;
3:  $\omega = \text{Random}()$ ;
4:  $\varepsilon = \text{CalcStepLength}(\tau, \omega)$ ;
5:  $T_k = \text{FindAdjacent}(T_i)$ ;
6:  $\varepsilon = \begin{cases} 1 & |\varepsilon| < 1 \\ |\varepsilon| & \text{otherwise} \end{cases}; \alpha' = \frac{\alpha}{\varepsilon}$ ;
7: for  $j = 0; j < T_k.\text{count}; j++$  do
8:    $P_{T_k[j]} = \text{CalcProbability}(T_i, T_k, j, \alpha')$ ;
9: end for
10:  $P_{T_k} = \sum_{j=0}^{T_k.\text{count}-1} P_{T_k[j]}$ ;
11: for  $j = 0; j < T_k.\text{count}; j++$  do
12:    $\Psi_{T_k[j]} = P_{T_k[j]} / P_{T_k}$ ;
13:    $\Phi_{T_k[j]} = \sum_{m=1}^j \Psi_{T_k[m]}$ ;
14: end for
15:  $\mu = \text{Random}(0, 1)$ 
16: for  $j = 0; j < T_k.\text{count}; j++$ 
17:   if  $\Phi_{T_k[j]} == \mu$  then
18:     return  $T_k[j]$ 
19:   end if
20: end for

```

that when selecting the next transaction from the predecessor transactions of W_{20} , the MR tends to select W_7 , which has a smaller ΔW . Once more, in Fig. 5(e), LFMR finally selects W_6 , which has a bigger ΔW . Finally, we obtained the transaction selection path shown in Fig. 5(f). Therefore, by introducing the Lévy flight algorithm, it is possible to dynamically adjust the selection tendency of MR, alleviating the problem of the nodes with bigger ΔW values not being selected for a long time. This results in a more uniform tip selection process.

The procedure for selecting the next transaction in the path using LFMR is shown in Algorithm 2. Where in this algorithm, α' is the new parameter calculated from α and ε , its value should be located in the range of 0 to 1, which leads to a more uniform selection of transactions; ε is the step size generated by the Lévy flight algorithm in the following equation:

$$\varepsilon = \frac{\tau}{|\omega|^{1/\theta}} \quad \tau \sim N(0, \varphi^2), \omega \sim N(0, 1) \quad (4)$$

where ε is a nonnegative variable whose value is random and follows the Lévy distribution. θ is a fixed number with a value range of 1 to 2. τ and ω are random numbers obedient to normal distribution. φ is the standard deviation of the normal distribution, it is calculated according to the following equation:

$$\varphi = \left[\frac{\Gamma(1 + \theta) * \sin((\theta/2) * \pi)}{\Gamma((1 + \theta)/2) * 2^{(\theta-1)/2} * \theta} \right]^{1/\theta} \quad (5)$$

where Γ represents the standard Gamma function and $\sin$ represents the standard sine function.

The probability that a particular transaction will be selected is $P_{T_k[j]}$, while the probability that all transactions in T_k will be selected is P_{T_k} . To determine which transaction will eventually be selected, a roulette algorithm is utilized. The $P_{T_k[j]}$ is regarded as the adaptability value in the roulette algorithm, $\Psi_{T_k[j]}$ indicates the probability that this part is selected and $\Phi_{T_k[j]}$ denotes the cumulative probability of this part. Depending on the range in which the random number μ settles, the final node is selected.

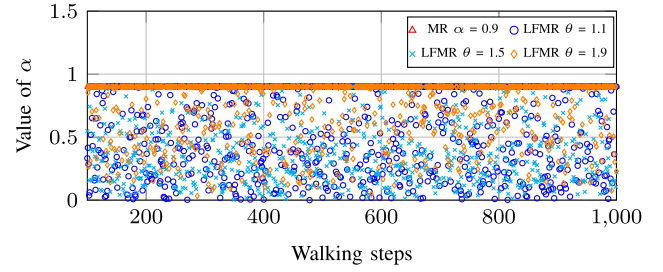


Fig. 6. Distribution of α values.

In Algorithm 2, lines 2, 3, and 15 calculate random numbers. Line 4 calculates the step length using (4). Line 5 fetches the reference to the relevant position of the T_i from the adjacency list of the DAG. Therefore, their time complexity is $O(n)$. Lines 7 to 9 utilize (3) in the loop to calculate $P_{T_k[j]}$. Line 10 involves the accumulation operation. Therefore, their time complexity is $O(n)$. Lines 11 to 14 contain a double loop, a for loop, and an accumulation operation. Thus, the time complexity is $O(n^2)$. Lines 16 to 20 contain one loop. Therefore, the time complexity is $O(n)$. In summary, the overall time complexity of this algorithm is $O(n^2)$. In this algorithm, storing the adjacent nodes of T_i , the probability, and the cumulative probability of the adjacent nodes is necessary. The space required is related to the number of adjacent nodes, resulting in a space complexity of $O(n)$.

IV. EXPERIMENTS

A. Tip Selection Algorithm Comparative Experiments 1

To assess the improvement of LFMR in TT selection speed, we design a program that performs triple iterations to generate experimental data.

- 1) The first loop generates the transaction numbers of the DAGs. The numbers range from 100 to 1000 in 100 increments.
- 2) The second loop iterates 50 times, each iteration randomly generating a DAG based on the number of transactions obtained in the first iteration. Each transaction will be assigned a weight of $3n$, and the value of n ranges from 0 to 4. Then, the program calculates the W for every transaction by adding the weights of its adjacent transactions. To guarantee the validity of the experimental data, each DAG contains at least 10 TTs.
- 3) The third loop also iterates 50 times. In each iteration, the MR and LFMR algorithms are employed to find TTs in the generated DAGs. The α value of MR is set to 0.9, and the θ values of LFMR are set to 1.1, 1.5, and 1.9. The number of walking steps required for the two algorithms to find two different TTs in each DAG, as well as the α values adjusted by LFMR, are recorded. Considering the possibility that two different TTs may not be found for a long time, we set the upper limit of the number of walking steps to 10 000. Any number exceeding 10 000 will be capped at 10 000.

As shown in Fig. 6, LFMR can effectively randomize the values of α . The effect of this adjustment is illustrated in Fig. 7. When the α of MR is set to 0.9, the number of walking steps frequently approaches or exceeds 10 000. This indicates that using the MR algorithm for random walks has a greater probability of

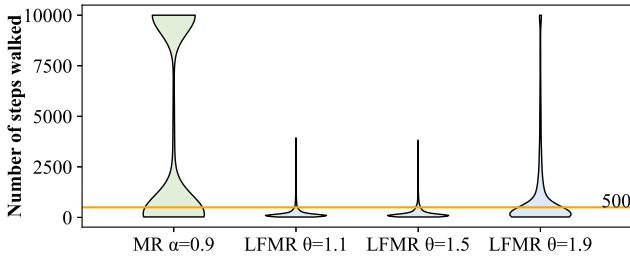


Fig. 7. Number of walking steps required for MR and LFMR to find two different TTs.

taking a long time to find two different TTs. Conversely, when using the LFMR algorithm with the θ values set to 1.1 and 1.5, the walking steps are generally concentrated within 500. For a θ value of 1.9, the walking steps are predominantly around 500, with only a few instances nearing or surpassing 10 000. Consequently, LFMR outpaces MR in the speed of tip selection.

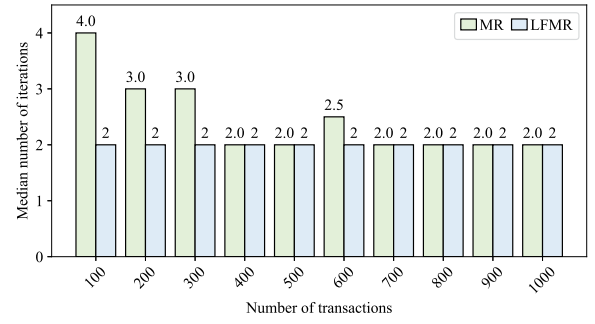
B. Tip Selection Algorithm Comparative Experiments 2

Furthermore, to verify the improvement of LFMR in tip selection stability, experiments are conducted for both MR and LFMR in DAGs with transaction numbers ranging from 100 to 1000. The MR runs 50 times with different α , and the LFMR runs 50 times with different θ . The values of α range from 0.1 to 0.9 in 0.1 increments. The values of θ range from 1.1 to 1.9 in 0.1 increments. The number of iterations required to find the two TTs is recorded. The results are shown in Fig. 8. Both the median and average numbers of iterations of the LFMR stabilize at 2. The standard variances of the iterations remain within 1. Thus, we can conclude that the number of iterations in the LFMR typically stabilizes around 2. In the MR, median numbers of iterations remain relatively stable, but average numbers and standard variances fluctuate widely. In summary, the LFMR can assist IOTA in achieving more efficient and stable device access authentication.

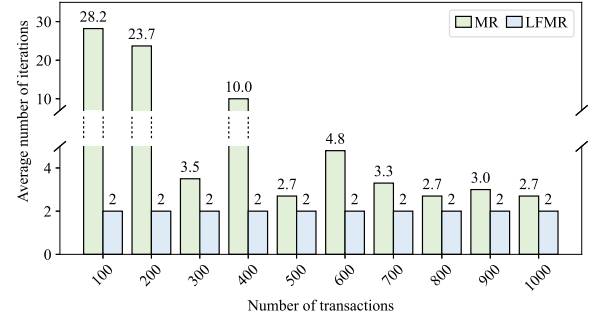
C. Tip Selection Algorithm Comparative Experiments 3

To assess the improvement of LFMR in transaction confirmation efficiency, we designed a program to compare the number of unconfirmed transactions over a certain period using MR and the proposed LFMR algorithms. The default value of α is set to 0.9, and θ is set to 1.5.

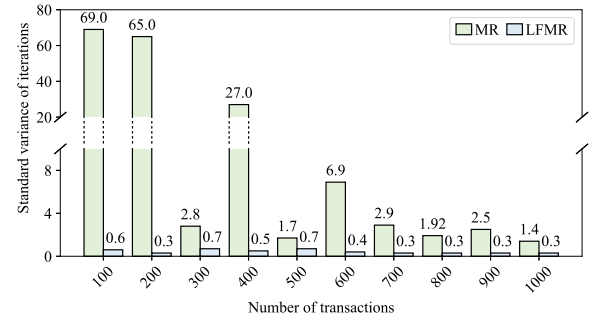
- 1) The first iteration sends transactions to the DAG at variable speeds. The speeds range from 100 to 1000 transactions per UT. UT contains the time of transaction construction and the time of selecting and confirming two tips.
- 2) The second iteration involves conducting the experiment 10 times to obtain experimental data and compute the average values.
- 3) The third iteration executes the experiment for 300 UT, consistently adding transactions to the DAG. Ultimately, it calculates the number of unconfirmed transactions surpassing 10 UT across various confidence levels (CLs). CL is the percentage of the number of TTs directly or indirectly pointing to a transaction with the total number of TTs. When the CL of a transaction surpasses the predefined threshold, the transaction is considered confirmed.



(a)



(b)



(c)

Fig. 8. Results for the number of iterations required to find two TTs. (a) Comparison of the median numbers of iterations. (b) Comparison of the average numbers of iterations. (c) Comparison of the standard variances of iterations.

In Fig. 9, we present the experimental results across four CLs. It is observed that with an increase in η , there is a corresponding decrease in the ξ for both MR and LFMR algorithms. Notably, the reduction in ξ for the LFMR algorithm is more pronounced as compared to the MR algorithm. This finding suggests that the LFMR algorithm facilitates a higher rate of transaction confirmation, thereby implying increased simplicity of device access authentication.

D. Performance Experiments

To verify the practicability of the mechanism, we adopt 10 Raspberry Pi 4B as EDs. The Raspberry Pi 4B features a 1.5 GHz CPU and 4 GB of RAM. Besides, we install the private Tangle on one desktop computer (3.0 GHz frequency and 6-core CPU, 16 GB RAM) and two IOTA nodes on two laptops (each laptop features a 2.6 GHz frequency and 6-core CPU and 16 GB of RAM). The Tangle (2-core CPU, 2 GB RAM) and IOTA nodes (each node features 2-core CPU, 2 GB RAM) are installed in virtual machine systems. The devices are connected in a local

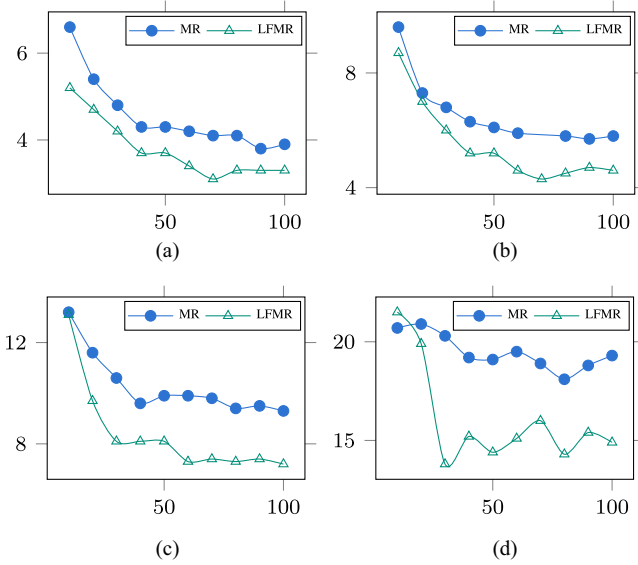


Fig. 9. Transaction confirmation efficiency experimental results. The horizontal axis represents the number of transactions sent per unit time (UT) (η). The vertical axis denotes the percentage of unconfirmed transactions to the total number of transactions (ξ). (a) $CL=0.2$ (b) $CL=0.4$ (c) $CL=0.6$ (d) $CL=0.8$.

area network environment. Thus, the influence of the network environment on test results can be eliminated.

In IIoT, we attach significant importance to signal response latency, so we first need to test the time overhead required to complete an IOTA transaction. To simulate the time required for device access when a large number of devices send messages to the Tangle simultaneously, we use multiple threads on the IOTA node in the virtual machine system to send messages to the Tangle concurrently. The Raspberry Pis contribute 40% of the message traffic, while laptops are responsible for the remaining 60%. For instance, at a rate of 100 messages per second, each Raspberry Pi sends 4 messages, and each laptop sends 30 messages. To evaluate the impact of varying communication pressures on the message transmission time and transaction confirmation time of ED access, we transmit messages from the IOTA node to the Tangle at different frequencies.

In addition, the minimum proof of work score (MPWS) value has an impact on both the security and complexity of the consensus algorithm in IOTA, higher values of MPWS take longer to execute. To determine the most appropriate MPWS value that can strike a balance between network performance and security, we conduct experiments to evaluate the time consumption of Raspberry Pi for executing the consensus algorithm under varying MPWS values.

Fig. 10(a) depicts the result of the two experiments. The legend of it represents the message-sending frequency, for example, “f-100” indicates 100 messages per second. Regarding MPWS, when the MPWS value remains within 10, the time required for the device to complete proof of work does not exceed 500 ms. Furthermore, from the perspective of different curves, our private tangle also exhibits good adaptability to high-frequency device access. The time overhead required to successfully send a transaction to the private tangle slightly increases from 200 to 350 ms with an increase in message transmission speed. The overall time overhead barely exceeds 400 ms. Considering these

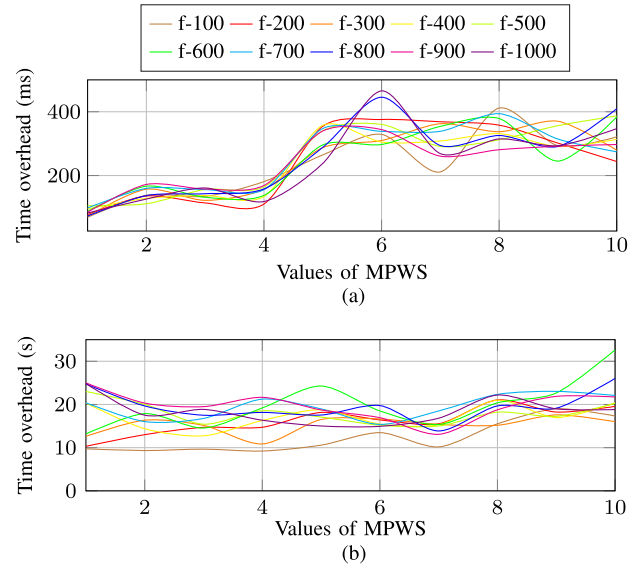


Fig. 10. Results of performance experiments. (a) Sending message time evaluation. (b) Message confirmation time evaluation.

two aspects, our solution fulfills the real-time requirements of IIoT device access authentication.

The result of the message confirmation time overhead measuring experiment is illustrated in Fig. 10b. The legend of it represents the message-sending frequency, for example, “f-100” indicates 100 messages per second. The overall confirmation time overhead for different message-sending frequencies and MPWSs is kept under 40 s, which is acceptable for device access.

Given that the IIoT scenario involves a significant number of devices that generate a large number of nodes in the Tangle, it is essential to consider the utilization of resources by the Tangle. A rapid increase of resource utilization by the Tangle as the number of devices in the IIoT scenario grows will significantly impact the operation of the entire system. Therefore, we designed an experiment to measure the resources used by the Tangle under different node quantities. In our experimental setup, we employ a multithreading approach to simulate multiple devices sending messages simultaneously while the devices send messages at a consistent rate. We conduct a series of simulations in our study, using a range of device quantities from 5000 to 50 000. Meanwhile, we employ a script to periodically capture the CPU, RAM, and disk usage of the system hosting the Tangle.

The experimental data in Table I show that the resources required for the Tangle to operate within our experimental environment are consistently at a low level. As the quantity of devices expands from 5000 to 50 000, the CPU utilization remains primarily within the threshold of 0.5%. Preceding the commencement of the experiment, the system occupies 1.29 GB of memory. Consequently, the Tangle utilizes less than 0.3 GB of memory and augments at a sluggish pace. Furthermore, the Tangle occupies less than 30 KB of disk space. As the quantity of nodes grows, the Tangle will temporarily accumulate data within an incremental snapshot file. Upon reaching a designated node threshold, the data will be compressed and merged into a full snapshot.

TABLE I
TANGLE RESOURCE UTILIZATION

Device number	CPU	Memory (GB)	Storage (B)
5000	0.54%	1.38	23 836
10 000	0.34%	1.39	23 836
15 000	0.79%	1.43	23 836
20 000	0.33%	1.47	26 172
25 000	0.23%	1.46	26 172
30 000	0.32%	1.47	26 172
35 000	0.38%	1.49	26 172
40 000	0.13%	1.49	26 172
45 000	0.42%	1.5	26 172
50 000	0.42%	1.51	26 172

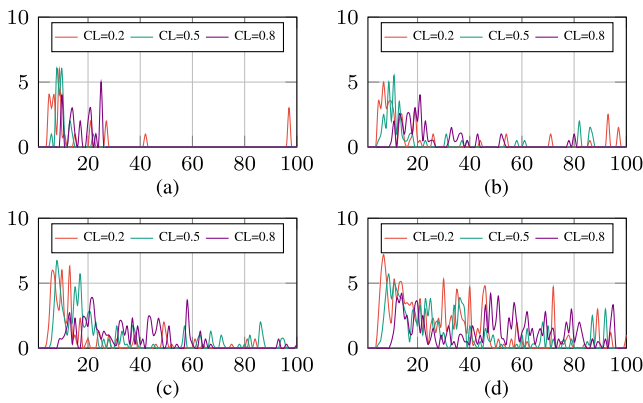


Fig. 11. Malicious transaction interception experimental results. In the four subfigures, the sending speed for legitimate transactions is uniformly 1000/UT, whereas those for malicious transactions vary. The y-axis in each subfigure is the attack confirmation rate(%). (a) 100/UT. (b) 200/UT. (c) 300/UT. (d) 400/UT.

E. Security Experiments

To validate the security efficacy of our proposed mechanism, we conducted experiments simulating the simultaneous generation of legitimate and malicious transactions. Legitimate transactions only confirm legitimate transactions, while malicious transactions will confirm both transactions. When a legitimate transaction selects a malicious TT, no processing is performed. In our experiment, legitimate transactions were sent at a consistent speed of 1000 per UT, with malicious transactions sent at varying speeds of 100, 200, 300, and 400 per UT. We evaluated the interception effect of malicious transactions by IOTA at CLs of 0.2, 0.5, and 0.8. Intercepting malicious transactions helps minimize the number of registration requests reaching the cloud platform.

Fig. 11 displays the experimental outcomes, with the attack confirmation rate reflecting the percentage of the number of malicious transactions confirmed to the number of malicious transactions sent at the current UT. From Fig. 11(a)–(d), the sending speed of legitimate transactions remains unchanged, and the sending speed of malicious transactions gradually increases. Across all subfigures, the attack confirmation rate exhibits a general downward trend. In Fig. 11(a), the attack confirmation rate approaches zero over time, exhibiting only minor fluctuations.

However, in Fig. 11(b)–(d), as the sending speed of malicious transactions increase, the fluctuations in the attack confirmation rate also increase, making it more difficult for the rate to approach zero. Thus, Fig. 11 illustrates that when the sending speed ratio of legitimate transactions to malicious transactions exceeds a certain value, the attack confirmation rate approaches zero over time. This means that, with a sufficient number of legitimate IIoT devices, the proposed mechanism effectively intercepts malicious transactions. Furthermore, Fig. 11(a)–(d) generally demonstrates that as the CL value increases, the trend of the lines shifts to the right, indicating that a larger CL value results in delayed attack confirmation. In Fig. 11(b)–(d), the varying CL values show a fluctuating effect on the attack confirmation rate without any obvious pattern. Fig. 11(a) reveals that from approximately the 30th UT, the attack is confirmed only when the CL value is 0.2, while the confirmation rate remains at zero for CL values of 0.5 and 0.8. This indicates that when the sending speed ratio of legitimate transactions to malicious transactions exceeds a certain value, a higher CL value can more effectively diminish the attack confirmation rate. Compared with centralized authentication, our proposed mechanism can intercept most attacks through IOTA, thereby alleviating the burden on the cloud platform in processing device access authentication.

V. CONCLUSION

In this article, a secure device access mechanism for IIoT is devised. The mechanism adopts blockchain technology to provide a device authentication service. We designed a device authentication model to verify the legitimacy of the EDs and FDs. An optimized tips selection algorithm is developed to achieve a more reasonable tips selection, thereby balancing device authentication wait time and security. The experiment results show the proposed mechanism is feasible in industrial scenarios.

In the future, to further enhance the effectiveness of this mechanism, some modification schemes will be proposed, including 1) integrating FPGA to accelerate the algorithms, and 2) adding IIOT-related factors to the proof-of-work algorithm.

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